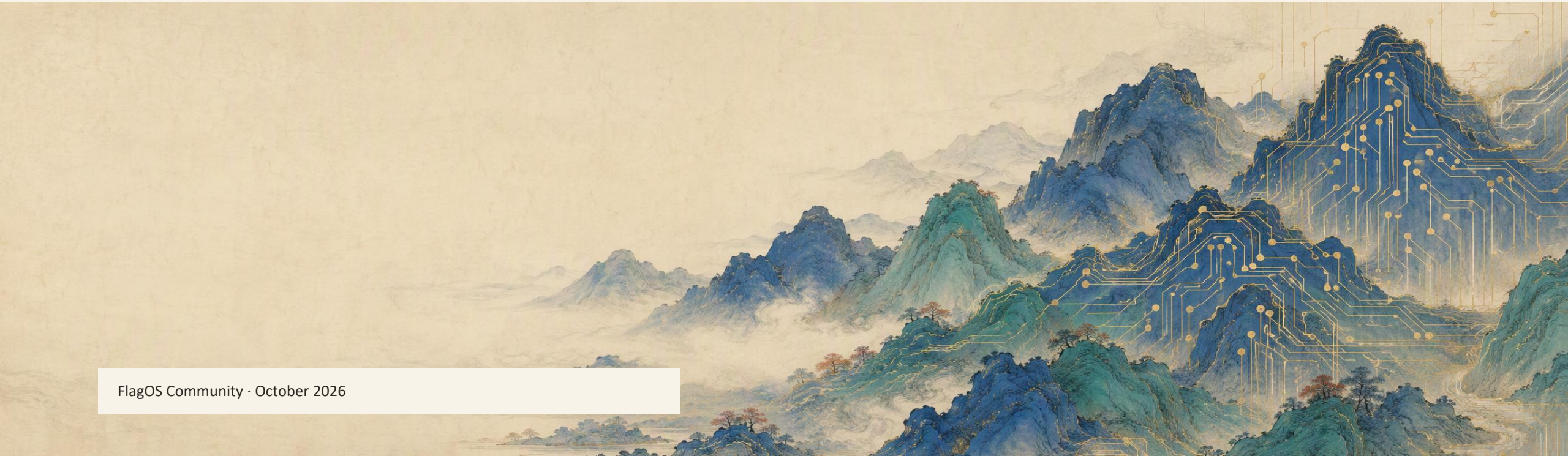
BEIJING ACADEMY OF ARTIFICIAL INTELLIGENCE 北京智源人工智能研究院

Global AI Compute Faculty Development Initiative

Open AI Computing
for Universal Benefit



FlagOS Community · October 2026

Taking Open Computing Global — Together

Standards & Community

ITU-T F.FUCL — the first international standard for unified communication across AI hardware.
ITU-T F.FUCL —— 首个面向 AI 硬件统一通信的国际标准。
Co-building with global open-source communities: Linux Foundation, IEEE, Eclipse, Hugging Face, PyTorch.
与全球开源社区共建：Linux 基金会、IEEE、Eclipse、Hugging Face、PyTorch。

The Global Initiative

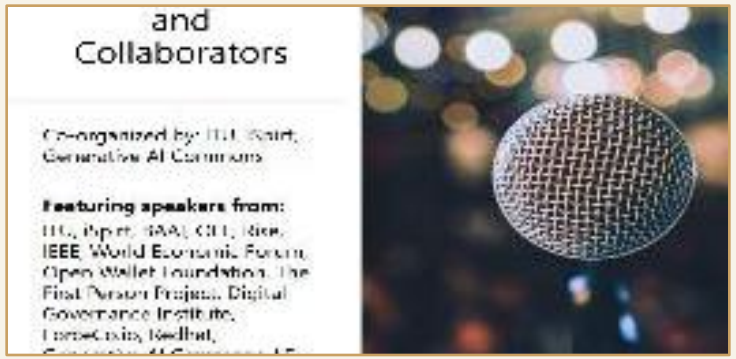
Global Open Compute Initiative with UN ODET, UNESCO, African Union ASRIC, Peking University, BAAI and the FlagOS community.
「全球开放计算倡议」由联合国数字与新兴技术办公室、联合国教科文组织、非盟 ASRIC、北京大学、智源研究院与 FlagOS 社区共同发起。
Its first program — the **China–Africa AI Talent Development Program** — is already running. See next pages.
首个项目「中非 AI 人才培养计划」已经启动，详见后页。

Open Dialogue Worldwide

Paris · May 2025 — Open-Source Strategy Forum
巴黎 · 2025 年 5 月 —— 开源战略论坛
Geneva · Jul 2025 — GDC2025, global speakers & collaborators
日内瓦 · 2025 年 7 月 —— GDC2025，全球讲者与合作伙伴
Hangzhou · Sep 2025 — Open SDG Conference, co-organized with the UN
杭州 · 2025 年 9 月 —— 与联合国共同举办的开放 SDG 大会



Open-Source Strategy Forum, Paris 开源战略论坛，巴黎

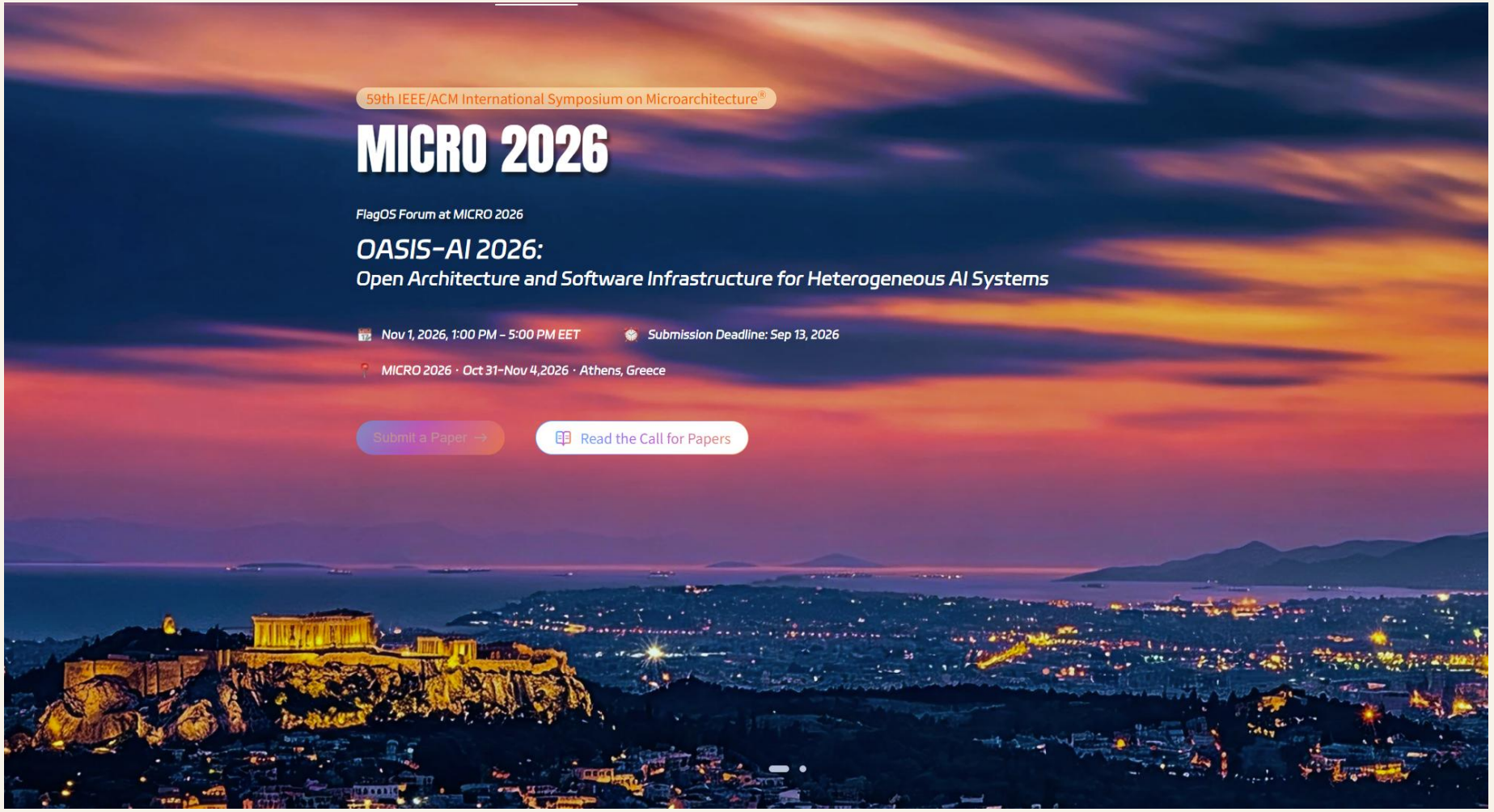


GDC2025, Geneva GDC2025，日内瓦



ITU AIforGood Award 联合国ITU 人工智能向善大奖

OASIS-AI 2026



Global Open Compute Initiative for AI Talent
面向 AI 人才的全球开放计算倡议

Open Compute turns underutilized AI compute resources worldwide into accessible infrastructure for developing regions — enabling more inclusive global AI innovation, starting with education.
开放计算把全球未被充分利用的 AI 算力，转化为发展中地区可获取的基础设施，让全球 AI 创新更具包容性 —— 从教育开始。

THE PARTNERS 合作伙伴

UN Office for Digital & Emerging Technologies
联合国数字与新兴技术办公室
UNESCO
联合国教科文组织
African Union — ASRIC
非洲联盟 —— ASRIC
Peking University · BAAI · FlagOS Community
北京大学 · 智源研究院 · FlagOS 社区

THE FIRST PROGRAM 首个项目

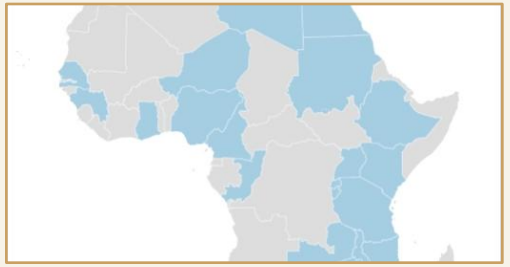
China–Africa AI Talent Development Program
中非 AI 人才培养计划

April 15–30, 2026 · Beijing. BAAI, AU-ASRIC and Peking University jointly welcomed **10+ university educators from 10 African countries** for hands-on training in AI systems and FlagOS practice.
2026 年 4 月 15—30 日 · 北京。智源研究院、非盟 ASRIC 与北京大学共同迎来来自 10 个非洲国家的 10 余位高校教师，开展 AI 系统与 FlagOS 实操培训。
Graduates returned home as certified trainers — multiplying the program's reach across the continent.
结业学员回国后成为认证讲师，让项目影响在非洲大陆持续放大。



Graduation ceremony, Beijing — April 2026 结业典礼，北京 —— 2026 年 4 月


Educators from 10 countries 来自 10 个国家的教师

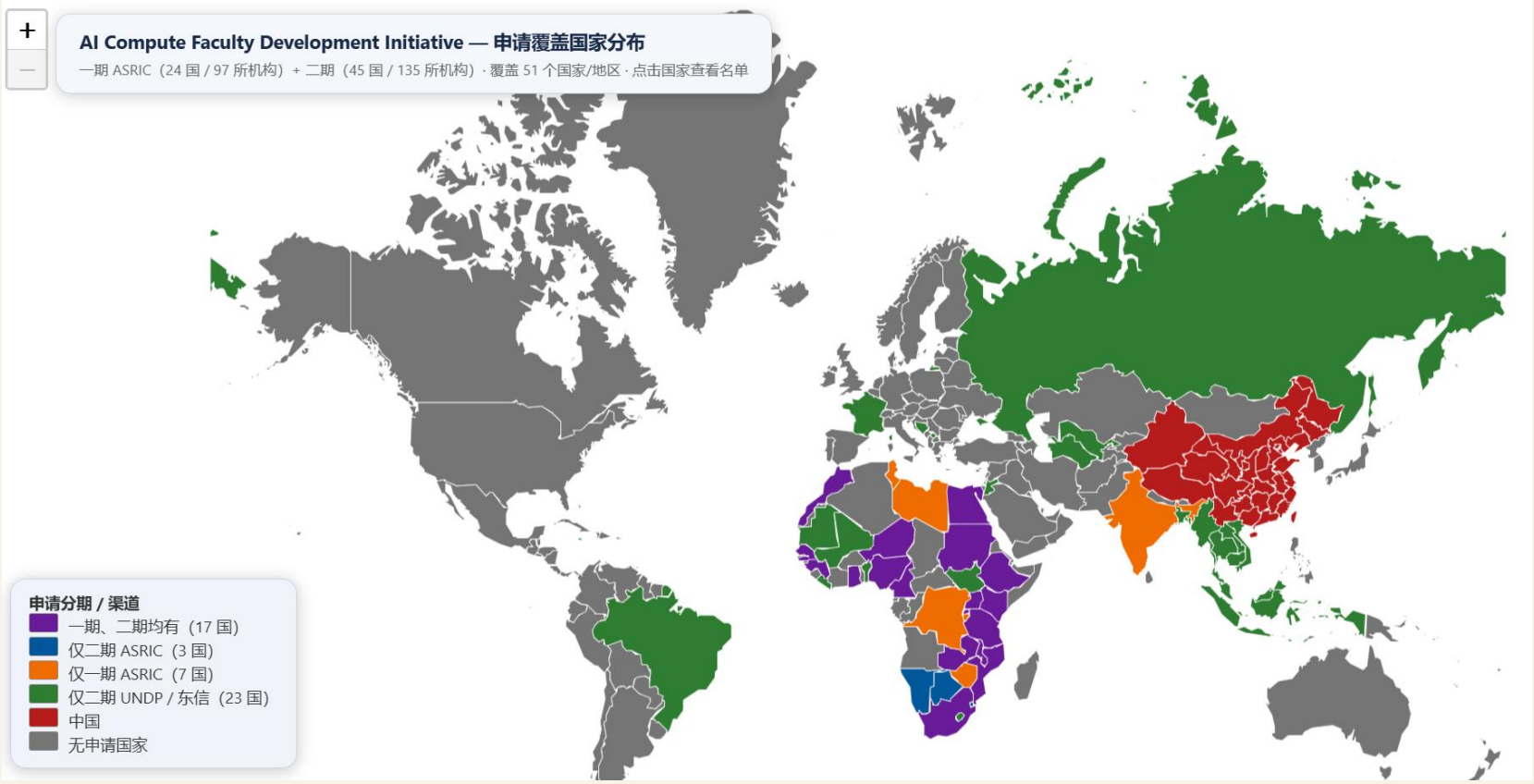

Participating countries across Africa 参与项目的非洲国家

2nd Cohort Global Open Compute Initiative for AI Talent
全球开放计算 AI 人才计划 · 第二期

Expanding from Africa to Asia, South America, Europe, and Beyond (October 2026)
从非洲拓展至亚洲、南美、欧洲及更多地区（2026 年 10 月）

In collaboration with the African Union–African Scientific, Research and Innovation Council, Peking University, the United Nations Development Programme, ASEAN and other partners, BAAI will launch the second cohort of Open Compute–based AI faculty training in October 2026. Faculty members from more than 50 countries and over 100 universities have already applied to participate.

智源研究院将携手非洲联盟—非洲科学研究与创新理事会、北京大学、联合国开发计划署、东盟等伙伴，于 2026 年 10 月启动第二期基于开放计算的 AI 师资培训。目前已有来自 50 多个国家、100 余所高校的教师报名参与。



Schedule

Day	Date	Module	AM (09:00–12:00)	PM (14:00–17:00)
1	Oct 12 (Mon)	Opening & Course Overview	◆ Opening Ceremony & Icebreaking • Welcome session • Workshop goals • Participant self-introductions • Course overview and lab platform introduction	◆ General Course on AI Large Models • Basics of LLM, inference, training • FlagOS software stack and inference pipeline ■ Inference Lab (Full Pipeline) • End-to-end inference: model download, vLLM launch, client queries • RAG setup with embedding model and retrieval • Comparison of Flash Attention toggles
2	Oct 13 (Tue)	Inference & LLM Fundamentals	◆ Edge Model Vendor & Edge Device • Edge AI chips: power–memory–compute trade-offs • Lightweight model design: quantization, pruning, distillation • Labs	■ Lab: Triton Primer • Triton programming model (grid/block/thread) • Vector addition kernel: write, compile, run • Memory coalescing and optimization
3	Oct 14 (Wed)	AI Systems & Edge Computing	■ Edge Device Labs • Edge AI chips: power–memory–compute trade-offs • Lightweight model design: quantization, pruning, distillation • Lab: edge device	▶ Module 1 Theory: AI Systems Fundamentals and Heterogeneous Computing • Heterogeneous SIMT architecture (thread hierarchy, memory hierarchy) • Roofline model: compute vs. bandwidth • Memory management and data transfer
4	Oct 15 (Thu)	Kernel Optimization & AI Talent Exchange	■ Lab: Fused Add + RMSNorm • Implement separate Add and RMSNorm kernels (baseline) • Implement fused kernel and compare performance • Analyze memory access reduction and speedup	◆ AI4Good Global Exchange Forum • AI Education Exchange • Brainstorming on AI Talent Cooperation Plan
5	Oct 16 (Fri)	High-Performance AI Operators & Company Visit	▶ Module 2 Theory: High-Performance AI Operators and Operator Engineering • PyTorch call chain: Python → C++ → CUDA → GPU • Attention mechanism and $O(n^2)$ complexity • Operator fusion principles and common patterns • FlagGems operator library architecture	◆ China Tech and Traditional Culture; AI Compute Course Overview • Introduction to Chinese Technology and Traditional Chinese Culture ■ Lab: Profiler Demo • Profiler tool: call graph and kernel-level timing • Demo on image generation (Stable Diffusion) as workload • Identify bottlenecks and apply optimizations • Compare pre- and post-optimization performance

Schedule

Oct 17-18
(Saturday-Sunday)



Schedule

8	Oct 19 (Mon)	Kernel Generation & Performance Benchmarking	■ KernelGen: Generated Operators • Principle: LLM-assisted code generation from natural language • Hands-on: generate Triton kernels automatically • Performance comparison: handwritten vs. generated ▶ Module 5 Theory: Performance Benchmarking and Next-Gen Kernel Generation • Roofline model and performance metrics • Auto-tuning (FlagPerf) and LLM-assisted programming	◆ Large Model Vendor Talk （1 hour） • AI application showcases • University–industry collaboration discussion ◆ “Open Source AI and Global Public Goods: Insights from UN Practice” by Serge Stinckwich, Former Director of UNU Macau (1 hour) ◆ Business Exchange for Haidian-based Enterprises (2 hours) ⏱ Extended session: 14:00–18:00 (total 4 hours)
9	Oct 20 (Tue)	Performance Profiling & AI Compilers	■ Company Visit 1: Tencent/huawei	▶ Module 3 Theory + Lab: AI Compiler Principles and Optimization ■ Lab: • Triton optimization pass (operator fusion, memory layout) ▶ Theory: • AI compilers (IR, SSA, MLIR, FlagTree) • FX Graph and Conv+BN fusion pass • Compiler optimizations: LICM, loop unrolling
10	Oct 21 (Wed)	Distributed Training & Parallelism	■ Lab: Distributed Training • FlagScale framework and 3D parallelism (data/tensor/pipeline) • Configure and launch multi-node training • Communication primitives (AllReduce, AllGather) • LLM fine-tuning with local corpus	▶ Module 4 Theory (PART 1/2): Distributed Parallel Training • Distributed training: DDP, Megatron-LM, 1F1B pipeline • Communication primitives (AllReduce/AllGather) • FlagCX abstraction over NCCL/RCCL
11	Oct 22 (Thu)	Distributed Training & Industry Visit	▶ Module 4 Theory (PART 2/2): Distributed Communication Primitives • Communication Primitives • Communication Topology Optimization	■ Company Visit 2: Xiaomi or Huawei • (Xiaomi) Edge AI in smart EVs • (Huawei) Ascend AI chip and ecosystem
12	Oct 23 (Fri)	Pedagogy & Closing	◆ Pedagogy Discussion • Agent-driven teaching • Roundtable: how to bring the course back home	◆ Closing Ceremony • Group presentations of learning outcomes • Certificate distribution and group photo • Closing address and networking
13–14	Oct 24–25 (Sat–Sun)	Weekend Break	Weekend Break (free time)	Weekend Break (free time)

China-Africa /1st Cohort

The training experience: strong content, warm welcome

Rigorous, well-sequenced curriculum: AI systems, operators, compilers, distributed training, performance analysis.

Distributed Parallel Training and Communication was the most clearly explained module.

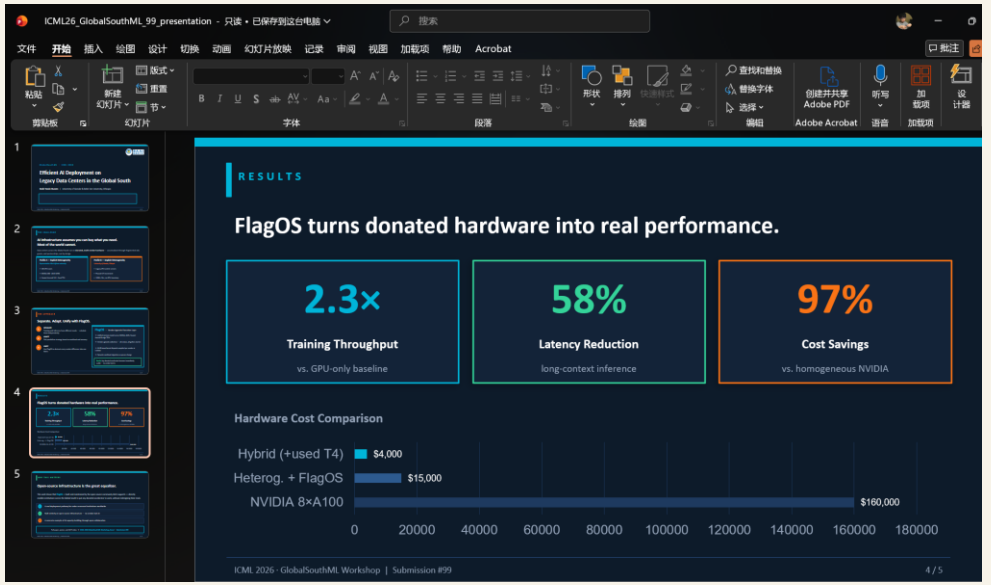
Excellent organization and a genuinely warm welcome from the BAAI team throughout the two weeks.

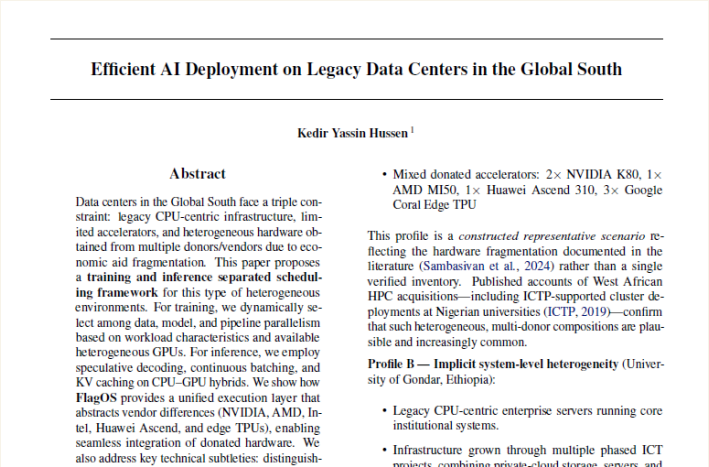
OVERALL RATING

Very Good

Balance of theory and hands-on labs, plus company visits and cultural exchange

Based on Phase 1 post-training evaluation





Efficient AI Deployment on Legacy Data Centers in the Global South

Kedir Yassin Hussen [1]

Abstract

Data centers in the Global South face a triple constraint: legacy CPU-centric infrastructure, limited accelerators, and heterogeneous hardware obtained from multiple donors/vendors due to economic aid fragmentation. This paper proposes a **training and inference separated scheduling framework** for this type of heterogeneous environments. For training, we dynamically select among data, model, and pipeline parallelism based on workload characteristics and available heterogeneous GPUs. For inference, we employ speculative decoding, continuous batching, and KV caching on CPU–GPU hybrids. We show how **FlagOS** provides a unified execution layer that abstracts vendor differences (NVIDIA, AMD, Intel, Huawei Ascend, and edge TPUs), enabling seamless integration of donated hardware. We also address key technical subtleties: distinguish-

- Mixed donated accelerators: 2× NVIDIA K80, 1× AMD MI50, 1× Huawei Ascend 310, 3× Google Coral Edge TPU

This profile is a *constructed representative scenario* reflecting the hardware fragmentation documented in the literature (Sambasivan et al., 2024) rather than a single verified inventory. Published accounts of West African HPC acquisitions—including ICTP-supported cluster deployments at Nigerian universities (ICTP, 2019)—confirm that such heterogeneous, multi-donor compositions are plausible and increasingly common.

Profile B — Implicit system-level heterogeneity (University of Gondar, Ethiopia):

- Legacy CPU-centric enterprise servers running core institutional systems.
- Infrastructure grown through multiple phased ICT

China-Africa /1st Cohort

AI courses and training activities at SUA / DIIT

Progress in introducing, implementing, or strengthening AI learning activities

1. **FlagOS practical orientation**
   Introduced access, workspace setup, and model-running activities to students and lecturers.



2. **Hands-on model execution**
   Practiced running AI models, observing outputs, managing experiments, and understanding compute needs.

3. **Awareness and peer learning**
   Built interest among students and lecturers while the team supported knowledge transfer.

4. **Teaching and research link**
   Connected FlagOS practice with applied AI work in agriculture, education, and data systems.

Outcome: stronger practical capacity to run AI models using FlagOS

AU-ASRIC / BAAI Follow-Up | SUA - DIIT 2/4



1 · PROGRESS REPORT

Mobilizing the University of Gondar

On return, the training was cascaded through every level of institutional leadership to build ownership before launching a public call.

Department → College → University Leadership

Outcome: leadership recognized the urgency of AI compute education for the university community.

Open Call for Applications

~400 applicants registered university-wide — far exceeding available training capacity.

Trainee Pool Segmented into 5 Groups

1. PhD / MSc candidates in urgent need of the course
2. PhD / MSc holders keen to co-teach the course
3. Graduating CS / IT / IS / InfoSci undergraduates
4. 3rd-year & below undergraduates
5. ICT staff managing the university data center

3

1 · PROGRESS REPORT

Six-Day Training — Delivered

60 priority trainees (mostly PhD/MSc students and ICT staff) completed a hands-on six-day program; soon-to-graduate undergraduates joined a one-day awareness track.

400+	60	1-day	60
University-wide applicants	Trainees completed the 6-day program	Awareness track for graduating & junior undergrads	Certificates of completion awarded

2 · COURSE INTEGRATION

Embedding AI Compute in the Curriculum

Partnerships & MoUs

- **University of Gondar – AU-ASRIC – BAAI**
  Tripartite Memorandum of Understanding signed, formalizing the collaboration.
- **Bahir Dar University**
  A second MoU is in progress, extending the initiative to a new partner institution.

Demand signal

Growing, unsolicited requests for compute access and mentorship from PhD researchers across disciplines and student start-up founders — demand now exceeds what informal handling can sustain.

Curriculum Roadmap

2025/26 (done)	Delivered as a stand-alone short-term training — too late in the academic year for formal curriculum entry.
Now	University legal office reviewing the MoU; course-integration proposal under active discussion.
2026/27 (planned)	Integration into the 4th-year, 1st-semester curriculum for AI-related programs.

China-Africa /1st Cohort





China-Africa /1st Cohort



China-Africa /1st Cohort



China's First Faculty Training Program on Autonomous AI Hardware & Software Technologies

66 Teachers Participating

34 Universities

12 Beijing Universities



- Built on FlagOS open-source technology
- 12 Beijing universities participated
- Open-sourced courses on domestic AI compute platforms

TextBook

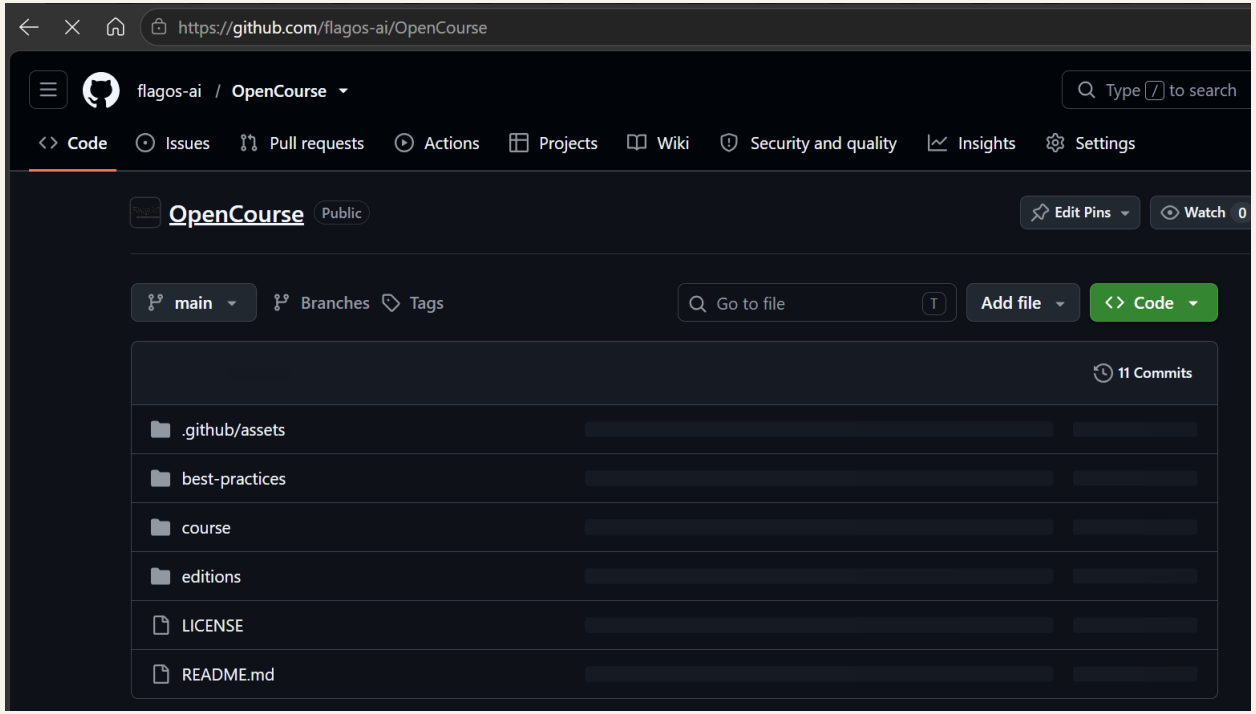

Global AI Compute Faculty
Development Program
Beijing 2026 | Supported by UNDP
Together, Let's Build an Inclusive AI Future
SCHEDULE
Manuel
Systèmes de calcul intelligent fondements logiciels et matériels
Module 04a : Entraînement parallèle distribué

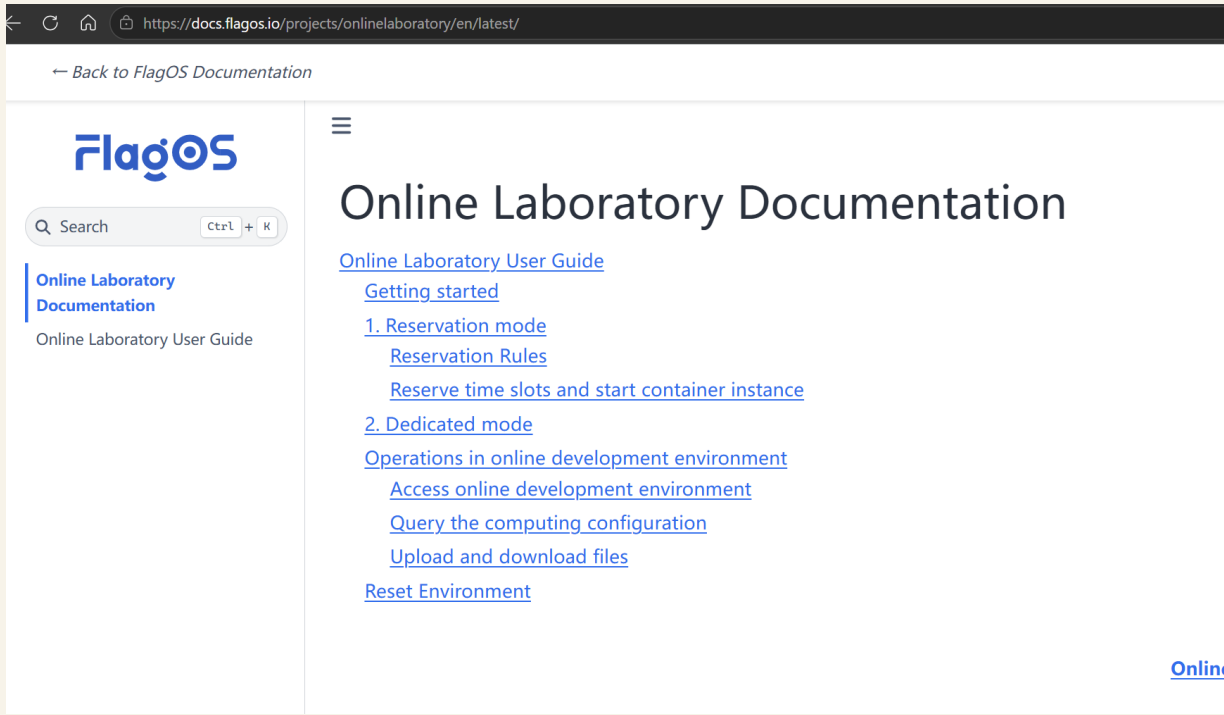

Global AI Compute Faculty
Development Program
Beijing 2026 | Supported by UNDP
TEXTBOOK


2026 Cohort
Confirmed Participants
Global AI Compute Faculty
Development Program
Beijing 2026 | Supported by UNDP


Global AI Compute Faculty
Development Program
Beijing 2026 | Supported by UNDP
TEXTBOOK

Open Course & online Lab





AI4Good Global Exchange · Agenda

Date & Time

October 15, 2026 | 14:00 – 17:00

Part 1: BAAI Introduction		
Time	Speaker	Topic
14:00 – 14:30	Lin Yonghua	BAAI Introduction: institutional overview, mission positioning and research directions
Part 2: AI4Good Co-Creation Forum		
14:30 – 15:10	Feng Zhengpeng / Liu Guang Beijing Academy of Artificial Intelligence (BAAI)	Introduction to the Earth Spatio-Temporal Large Model and Its Global Co-Creation
15:10 – 15:50	Co-founder ModelBest	Exploring a Global Co-Creation Model for Sovereign Large Models
15:50 – 16:30	Zhou Jie / Zhao Shuai Cixi Technology / BAAI	Edge AI for Education & Innovation
Part 3: Open Discussion		
16:30 – 17:00	All Participants	Open Discussion: focused on AI frontier technologies, large models, FlagOS and more

Enterprise Visit

HuaWei
20th Oct



Xiaomi

22nd Oct



Suggested tourist attractions in Beijing

24 Oct: forbidden City



17-18 Oct: Summer Palace



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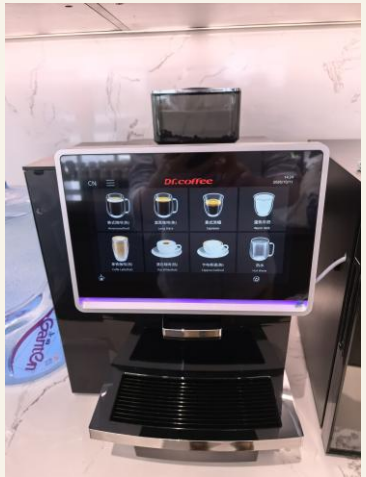
Class Placement Arrangement

Country	Region	Selection Channel	Professor	Class	Mentor	Coordinator	Lab Group
Ethiopia	Africa	African Union	Gebremichael Kibret Sheferaw	A	Teacher Luo	Zhao Zhiqiang	1
Botswana	Africa	UNDP	Dr Moemedi Lefoane	A	Teacher Luo	Zhao Zhiqiang	1
Ghana	Africa	UNDP	Prof. Delali Kwasi Dake	A	Teacher Luo	Zhao Zhiqiang	2
Lesotho	Africa	UNDP	Mohau Mamotuba Cynthia Nei	A	Teacher Luo	Zhao Zhiqiang	2
Rwanda	Africa	UNDP	Prof. Hanyurwimfura Damien	A	Teacher Luo	Zhao Zhiqiang	3
Malawi	Africa	UNDP	Assoc. Prof. Bennett Kankuzi	A	Teacher Luo	Zhao Zhiqiang	3
Namibia	Africa	African Union	Maria Ndapewa Ntinda	A	Teacher Luo	Zhao Zhiqiang	4
South Africa	Africa	UNDP	Mr. Samuel Viwe Mqaqa	A	Teacher Luo	Zhao Zhiqiang	4
Niger	Africa	African Union	Alio Boubacar Goga	A	Teacher Luo	Zhao Zhiqiang	5
Uganda	Africa	African Union	Kassim Kalinaki	A	Teacher Luo	Zhao Zhiqiang	5
Zambia	Africa	UNDP	Dr. Lighton Phiri	A	Teacher Luo	Zhao Zhiqiang	6
Bosnia and Herzegovina	Europe	UNDP	Sead Delalić	A	Teacher Luo	Zhao Zhiqiang	6
Cambodia	Asia	UNDP	Porchourng CHUOR	A	Teacher Luo	Zhao Zhiqiang	7
Malaysia	Asia	Eastcom	Lum Kin Yun	A	Teacher Luo	Zhao Zhiqiang	7
Myanmar	Asia	Eastcom	Aung Myat Khaing	A	Teacher Luo	Zhao Zhiqiang	8
Thailand	Asia	Eastcom	Suwichaya Suwanwimolkul	A	Teacher Luo	Zhao Zhiqiang	8
Uzbekistan	Asia	UNDP	Shirin Primkulova	A	Teacher Luo	Zhao Zhiqiang	9
Jordan	Asia	UNDP	Assoc. Prof. Dr. Esra’a Alshdaifat (Esra’a Ahmad Alshdaifat)	A	Teacher Luo	Zhao Zhiqiang	9
Vietnam	Asia	Eastcom	Nguyen Dinh Tran Long	A	Teacher Luo	Zhao Zhiqiang	10
Vietnam	Asia	Eastcom	Bui Thi Quynh Anh	A	Teacher Luo	Zhao Zhiqiang	10
Ethiopia	Africa	African Union	Yirga Yayeh Munaye	A	Teacher Luo	Zhao Zhiqiang	11
Nigeria	Africa	African Union	Arome Junior Gabriel	A	Teacher Luo	Zhao Zhiqiang	11
Ethiopia	Africa	UNDP	Dr. Elefelious Getachew Belay	B	Teacher Wang	Li Cheng	1
Guinea	Africa	UNDP	Dr. Ibrahima Kalil TOURE	B	Teacher Wang	Li Cheng	1
Kenya	Africa	UNDP	Prof. Joseph Muliaro Wafula	B	Teacher Wang	Li Cheng	2
Liberia	Africa	UNDP	Assoc. Prof. Dr. Darren Wilkins	B	Teacher Wang	Li Cheng	2
Malawi	Africa	African Union	Chimango Nyasulu	B	Teacher Wang	Li Cheng	3
Mauritania	Africa	UNDP	Prof. Mohamed Mahmoud EL BENANY	B	Teacher Wang	Li Cheng	3
South Africa	Africa	African Union	Katleho Moloi	B	Teacher Wang	Li Cheng	4
South Sudan	Africa	UNDP	Thon Malek Garang	B	Teacher Wang	Li Cheng	4
Tanzania	Africa	African Union	Eunice David Likotiko	B	Teacher Wang	Li Cheng	5
Tanzania	Africa	UNDP	Dr. Ally Selemani Nyamawe	B	Teacher Wang	Li Cheng	5
Uganda	Africa	UNDP	Assoc. Prof. Businge Phelix Mbabazi	B	Teacher Wang	Li Cheng	6
Brazil	South America	UNDP	Prof. Cleber Zanchettin	B	Teacher Wang	Li Cheng	6
Russia	Europe/Asia	UNDP	BELCHUSOV Anatolij Aleksandrovich	B	Teacher Wang	Li Cheng	7
Laos	Asia	Eastcom	Mekvongsa SIYOTHA	B	Teacher Wang	Li Cheng	7
Bangladesh	Asia	UNDP	Dr. Chowdhury Saima Ferdous	B	Teacher Wang	Li Cheng	8
Thailand	Asia	Eastcom	Kanthanat Meetae	B	Teacher Wang	Li Cheng	8
Thailand	Asia	UNDP	Asst. Prof. Dr Dittaya Wanvarie	B	Teacher Wang	Li Cheng	9
Indonesia	Asia	Eastcom	Maryamah	B	Teacher Wang	Li Cheng	9
Vietnam	Asia	Eastcom	Khuong Thi Thu Hai	B	Teacher Wang	Li Cheng	10
Vietnam	Asia	UNDP	Nguyen Dinh Quan (Quan D. Nguyen)	B	Teacher Wang	Li Cheng	10
Mozambique	Africa	African Union	Celestino Vaz Tomás Jone Joanguete	B	Teacher Wang	Li Cheng	11

Pantry/restroom/&others

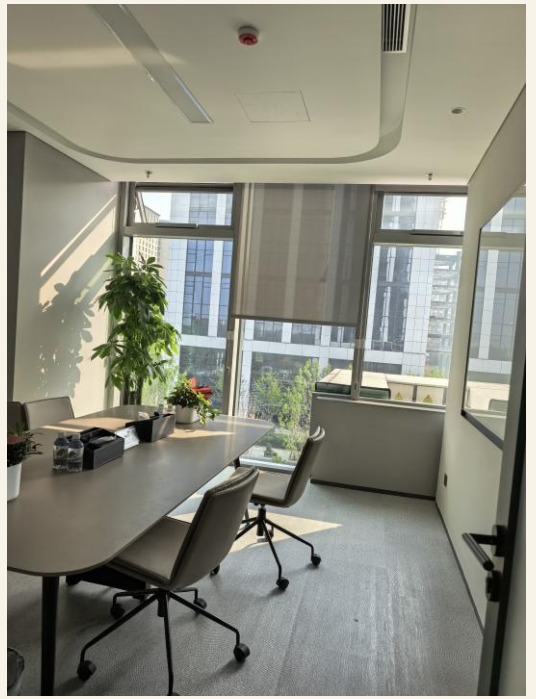
Pantry







muṣallā Room： 303



Breakfast & Lunch & Dinner

Complimentary services:

1. Breakfast is included in the hotel's charges.

2. Lunch is provided on each training day through pre-ordered takeout delivered to the classroom

The part to be paid by oneself:

1. Dinner
   You can explore Chinese cuisine with your friends, but you will need to pay for this part yourself.

2. Bar
   The hotel has bar itself.

If you have any dietary restrictions or preferences, please let us know.

Support Team



群聊：AI Compute Faculty Development Initiative

Serial number	unit	Name	role
1	BAAI (Beijing Zhiyuan Artificial Intelligence Research Institute)	Li Cheng	Head of Class B
2		Zhao Zhiqiang	Head of Class A
3		Lu Yijie	master control
4		Zhu Qigang	Project Leader
5		Yue Jinhui	assistant
6		Wang Yunbo	assistant
7	Peking University	Luo Guojie	Class A's main lecturer
8		Sun Xueting	Teaching Assistant
9		Wang Miaojie	Teaching Assistant
10		Long Yihan	Teaching Assistant
11		Li Haoran	Teaching Assistant
12		Kuang Daikang	Teaching Assistant
13	Renmin University of China	Wang Jing	Class B lecturer
14		Wang Yihang	Teaching Assistant
15		Cai Leyi	Teaching Assistant
16		Ye Jixuan	Teaching Assistant
17		Li Gan	Teaching Assistant
18		Xia Weihan	Teaching Assistant

WiFi/Free Internet Connection

Training Room

SSID： ZhongZhiYuan
Password： 12345678



Wifi is also available in your hotel for Free

Precautions

Be sure to abide by Chinese laws and respect Chinese cultural traditions

Subway:
Alipay/WeChat/visa/master/unionPay

We have prepared insurance for everyone, but still, safety should be taken into account：

For outpatient and hospitalization caused by accidents, compensation of 80,000 yuan can be provided.

For outpatient and hospitalization caused by sudden acute diseases, compensation of 80,000 yuan can be provided.

For sudden accidents, an ambulance can be called at a cost of 1,500 yuan

Resource

Code： https://github.com/flagos-ai
Open course： https://github.com/flagos-ai/OpenCourse
Online lab： https://flagos.io/OnlineLab
FlagOS official website： https://flagos.io
Wechat channel： FlagOpen
e-learning center： https://learnbuddy.tencent.com

Enjoy your stay in China
For
Every Day

AN OPEN INVITATION

Co-create an Open and Shared AI Ecosystem

Open computing turns underused AI compute everywhere into shared infrastructure for all — a huge opportunity, and a mission we share with every nation.



Adopt open standards · Open your compute · Bring AI to every classroom